STRIVE

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NAD+

Nicotinamide adenine dinucleotide (NAD+) is a vital cofactor present in all living cells, essential for a multitude of metabolic processes. It functions primarily as a carrier of electrons in redox reactions, crucial for the production and transfer of energy in cells. NAD+ is involved in key metabolic pathways such as glycolysis, the citric acid cycle, and oxidative phosphorylation.[1,2]

Beyond its metabolic roles, NAD+ is fundamental for DNA repair, cellular signaling, and gene expression regulation through its interaction with enzymes like sirtuins and poly (ADP-ribose) polymerases. The decline in NAD+ levels with age is associated with various age-related diseases, highlighting its potential in interventions aimed at enhancing health and longevity.[1,2]

Potential Benefits:[1-3]

- Promotes healthy brain function
- Slows cognitive decline
- Decreases chronic fatigue and boosts energy levels
- Regenerates the cells and enhances metabolism
- Slows the aging process
- Reduces body inflammation

Key Components:

- **Mitochondrial Energy Production:** NAD+ is an essential coenzyme in mitochondrial respiration, where it participates in the transfer of electrons during the electron transport chain. This process generates ATP, the primary energy currency of cells. NAD+ acts as a crucial mediator in the conversion of nutrients into energy within the mitochondria. Maintaining optimal NAD+ levels is essential for efficient energy production and overall mitochondrial function.[3]

- **Mitochondrial Metabolism:** NAD+ acts as a cofactor for enzymes that facilitate these processes. Adequate NAD+ levels are necessary for these metabolic pathways to function, supporting the production of energy and essential cellular building blocks.[3]

- **Neuroprotection:** NAD+ has been implicated in various mechanisms of neuroprotection. It plays a role in maintaining the health and function of neurons, including the repair of damaged DNA in neurons, and promoting the production of neuroprotective factors. Some research suggests that boosting NAD+ levels may have neuroprotective benefits and help mitigate the impact of neurodegenerative diseases.[2]

- **Aging:** NAD+ has garnered significant attention in the field of aging research. It is involved in regulating various processes associated with aging. NAD+ levels tend to decline with age, and this decline is thought to contribute to the aging process. Increasing levels of NAD+ can potentially slow the aging process.[1]

How Supplied:

NAD+ 200 mg/mL vial Injectable (503B)

- **Vial Size:** 6 mL and 10 mL
- **Storage:** Refrigerate [36°F to 46°F (2°C to 8°C)]
- **Beyond-Use Date:** 82 days

NAD+ 200 mg sublingual flex-dose tablet

- **Storage:** Refrigerate [36°F to 46°F (2°C to 8°C)]
- **Beyond-Use Date:** 180 days

Suggested Dosing:

Subcutaneous or Intramuscular

- Inject 10 units (0.1 mL) subcutaneously or intramuscularly three times weekly for 1 week
- Inject 20 units (0.2 mL) three times weekly for 1 week
- Inject 30 units (0.3 mL) three times weekly for 1 week
- Inject 40 units (0.4 mL) three times weekly for 1 week
- Inject 50 units (0.5 mL) three times weekly thereafter

Titrate to an individually targeted dose, not to exceed 50 units (0.5 mL) three times weekly. A higher starting dose may be appropriate for patients who are not NAD+ naive. Administer in the morning on an empty stomach.

Intravenous

- **Suggested Dosing for Potential Aging Benefits:** Two hour infusion of 250 mg of NAD+, utilizing a minimum of 500 mL Normal Saline
- **Suggested Dosing for Neurodegenerative Disorders:** Four hour infusion of 500 mg of NAD+, utilizing a minimum of 500 mL Normal Saline

Start low, go slow

Oral Flex-Dose Tablet:

Dissolve one-fourth to one tablet sublingually or buccally daily in the morning on an empty stomach.

Potential Side Effects:

Headache, nausea, and dizziness. Upon injection, muscle tightness (including chest tightness) may occur for a moment. Injection site reactions, including redness, tenderness, or swelling are possible.

Warnings:

Increased levels of NAD+ can enhance glycolysis, providing cancer cells with the energy needed for rapid proliferation. While NAD+ plays a critical role in cellular function and energy metabolism, its potential impact on cancer growth should be carefully considered. Providers and patients should exercise caution, particularly in individuals with a history of or susceptibility to cancer. Consult a healthcare professional before using NAD+ therapies.

References:

1. Conlon NJ. The Role of NAD+ in Regenerative Medicine. Plast Reconstr Surg. 2022 Oct 1;150(4 Suppl):41S-48S. doi: 10.1097/PRS.0000000000009673. Epub 2021 Sep 28. PMID: 36170435; PMCID: PMC9512238.
2. Braidy N, Berg J, Clement J, Khorshidi F, Poljak A, Jayasena T, Grant R, Sachdev P. Role of Nicotinamide Adenine Dinucleotide and Related Precursors as Therapeutic Targets for Age-Related Degenerative Diseases: Rationale, Biochemistry, Pharmacokinetics, and Outcomes. Antioxid Redox Signal. 2019 Jan 10;30(2):251-294. doi: 10.1089/ars.2017.7269. Epub 2018 May 11. PMID: 29634344; PMCID: PMC6277084.
3. Cantó C, Menzies KJ, Auwerx J. NAD(+) Metabolism and the Control of Energy Homeostasis: A Balancing Act between Mitochondria and the Nucleus. Cell Metab. 2015 Jul 7;22(1):31-53. doi: 10.1016/j.cmet.2015.05.023. Epub 2015 Jun 25. PMID: 26118927; PMCID: PMC4487780.